



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2023-24/EM-CHP/543526

Date:

05/10/2023(19:13:28)

Subject:REVISION OF LMI/ OGN/OPS/SYST/003 HANDLING OF
ELECTRICAL APPARATUS IN DISTRESS

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

No changes is envisaged by COS. Only reviewing is being done as advised in Technical Audit.

May please approve.

Submitted by:

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Note No:2

Plz replace the doc file with pdf

Submitted by:

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On:07/10/2023(10:22:58) **Ref:**SMTPP/ELECT MAINT/2023-24/EM-CHP/543526

Note No:3

pdf file attached

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Note No:4

Reviewed.

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Note No:5

LMI reviewed.

May be approved

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On:09/10/2023(19:04:58) **Ref:**SMTTPP/ELECT MAINT/2023-24/EM-CHP/543526

Note No:6

LMI reviewed and revised Oct23, check list enclosed.

Submitted by:

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On:09/10/2023(19:58:43) **Ref:**SMTTPP/ELECT MAINT/2023-24/EM-CHP/543526

Note No:7

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On:10/10/2023(10:01:49) **Ref:**SMTTPP/ELECT MAINT/2023-24/EM-CHP/543526

Note No:8

It is requested to change Revision no. in LMI every where. (if LMI is changed). Superseded document no. & Record of Revision are to be mentioned

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On:10/10/2023(10:58:09) **Ref:**SMTTPP/ELECT MAINT/2023-24/EM-CHP/543526

Note No:9

Incorporate the changes suggested by EEMG.

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On:10/10/2023(13:18:16) **Ref:**SMTTPP/ELECT MAINT/2023-24/EM-CHP/543526

Note No:10

Suggestions by EEMG is incorporated.

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Note No:11

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On:10/10/2023(18:15:05) **Ref:**SMTTP/ELECT MAINT/2023-24/EM-CHP/543526

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On:16/10/2023(17:00:28) **Ref:**SMTTP/ELECT MAINT/2023-24/EM-CHP/543526

Note No:14

Submitted by:

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Note No:15

Approved & Submitted by:

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On:19/10/2023(11:30:33) **Ref:**SMTPP/ELECT MAINT/2023-24/EM-CHP/543526

NTPC Limited
SIMHADRI THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION
LMI/ OGN/OPS/SYST/003

Issue No: 2, Rev: 3 Date: October 2023

Handling of Electrical Apparatus in Distress

All locations are required to comply with this Directive

Approved for
Implementation by.....

Group General Manager (Simhadri)

Date:

NTPC Limited

SIMHADRI THERMAL POWER STATION

ELECTRICAL MAINTENANCE DEPARTMENT

LMI	Handling of Electrical Apparatus in Distress		
LMI No	LMI/OGN/OPS/SYST/003		
ISSUE No	2	REVISION: 3	DATE: October 2023

	DESIGNATION	SIGNATURE & NAME
Prepared by	DGM (EM)	
Checked by	DGM (EM)	
Reviewed by	AGM (EM) I/C	
Reviewed by	GM(Maint)	
Reviewed by	GM(O&M)	

Handling of Electrical Apparatus in Distress

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Handling of Electrical Apparatus in Distress

1.0 OBJECTIVE:

This LMI is being issued to comply with operation guidance note OGN/OPS/SYST/003, Rev 1 dated May 2020. The purpose of this LMI is to specify actions to be taken while handling electrical apparatus in distress. Operating and maintenance personnel should be aware of correct procedure of handling electrical apparatus in distress, in order to minimize risk to their life.

2.0 SCOPE:

Any electrical apparatus having a working voltage of 125V or higher, or where the fault energy could exceed the let through value of a 63 amp fuse-link.

3.0 REFERENCE DOCUMENT:

OPERATION GUIDANCE NOTE – OGN/OPS/SYST/003

4.0 SUPERSEDED DOCUMENT: Issue No:2,Rev:2 Date: May 2022

5.0 REVISION DETAILS: Rev2 to Rev-3

6.0 FAULTS ON ELECTRICAL APPARATUS / SWITCHGEAR:

The cause for initiation of fault in any electrical apparatus can be either looseness in power connection or broken or deteriorating component or ingress of moisture, or corrosion or contamination by foreign objects.

The fault once initiated normally results in overheating or burning of insulation. Burning of insulation generates a mixture of gases depending on the constituent of insulating material. Some of these fault gases are toxic and / or flammable. If the fault generates high temperature, the flammable gasses may ignite to cause fire or explosion.

Fault occurring in any operating switchgear may result in fire or explosion due to ignition of flammable gases while opening of power contacts of faulty circuit. Alternately the faulty circuit may not get isolated in time resulting in a bus bar fault or flashover.

7.0 IDENTIFICATION OF ELECTRICAL APPARATUS IN DISTRESS:

An Electrical apparatus/switchgear in distress can be identified by one or more of the following-

- i. Smoke, discolouration of paint, distortion of housing.
- ii. Sound of arcing or discharge.
- iii. Smell of burning insulation, varnish, paint etc.
- iv. Non-contact condition monitoring tools e.g. Partial discharge measurement, Thermography, *infrared temperature sensor*; motor current signature analyzer may be utilized for better identification and evaluation.

Operating/maintenance staff should be aware of the normal conditions of the apparatus including normal sounds while in operation. This all helps them in quickly identifying faulty conditions.

8.0 SAFETY INSTRUCTIONS(RECOMMENDATIONS):

Following safety instructions are to be followed while handling electrical apparatus in distress.

- i) When electrical apparatus is showing signs of distress, personnel should not approach the apparatus for investigation
- ii) Electrical apparatus showing signs of distress should not be operated locally.
- iii) Access to the area where the apparatus in distress is located should be prevented.
- iv) The suspected apparatus should be made dead from a remote point.
- v) The apparatus once isolated should be thoroughly examined, tested and faulty components replaced before returning the same to service.

9.0 THE RESPONSIBILITY CENTERS FOR IMPLEMENTATION:

Basic responsibility lies with respective Area in-charge.

10.0 RECORD KEEPING:

Any deviation shall be reported to HOD (EM).

11.0 REVIEW: Document will be reviewed once in two years by HOP (Simhadri).

12.0 CHECKLIST

An Electrical apparatus/switchgear in distress can be identified by one or more of the following-

- i. Smoke, discoloration of paint, distortion of housing.*
- ii. Sound of arcing or discharge.*
- iii. Smell of burning insulation, varnish, paint etc.*

Sl No	Description	Complied/not	Remarks
1	<i>Electrical apparatus showing signs of distress should not be operated locally</i>		
2	<i>Access to the area where the apparatus in distress is located should be prevented</i>		
3	<i>When electrical apparatus is showing signs of distress, personnel should not approach the apparatus for investigation</i>		
4	<i>The suspected apparatus should be made dead from a remote point</i>		
5	<i>The apparatus once isolated should be thoroughly examined. Tested faulty components replaced before returning the same to service</i>		